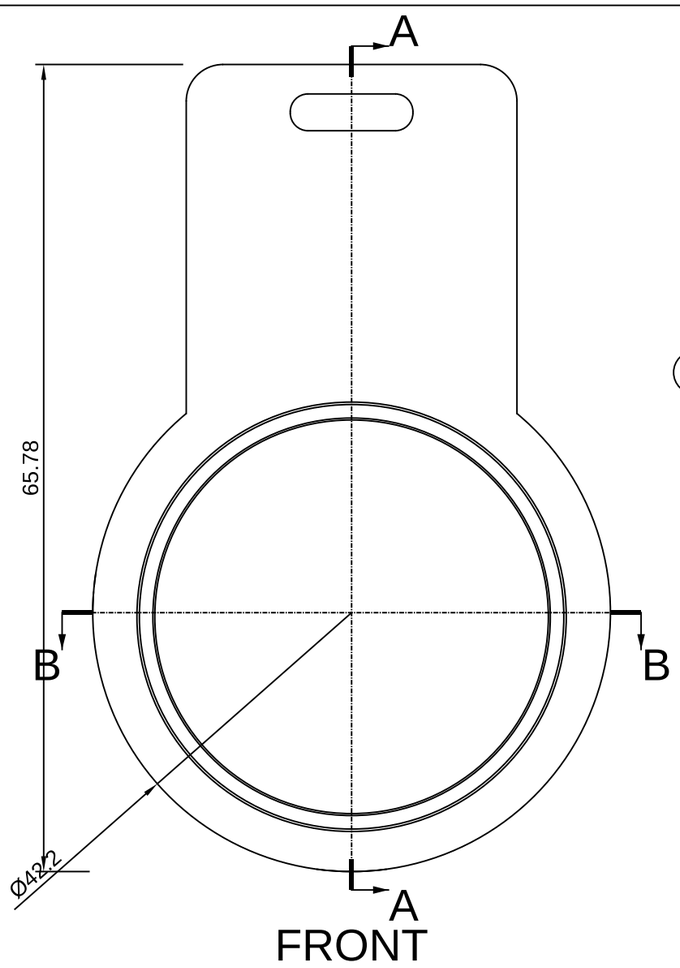
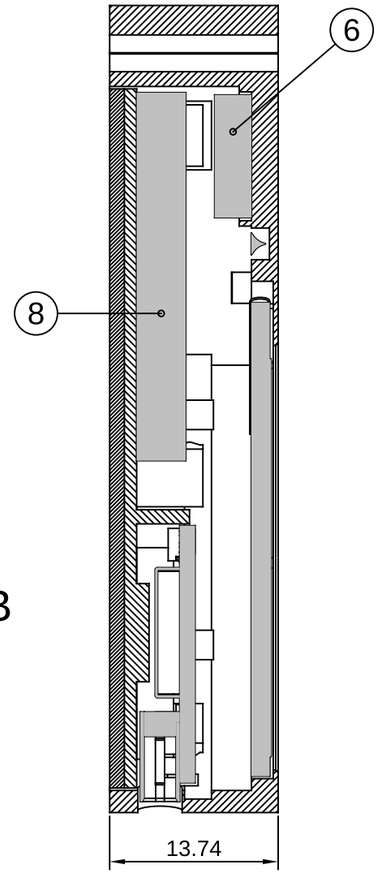
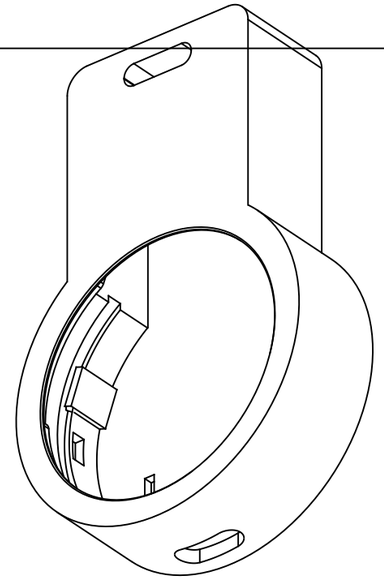


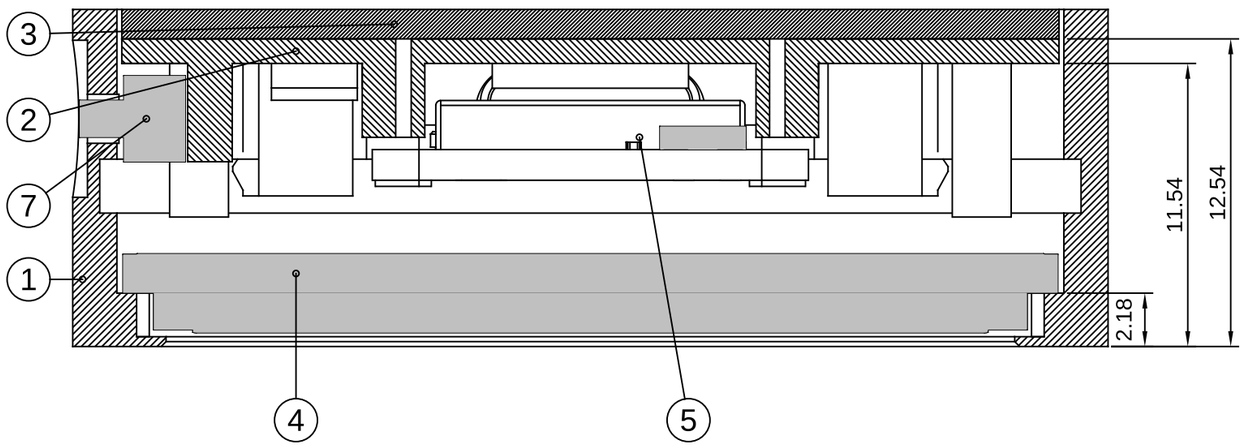


SECTION A-A



ISOMETRIC (1.4:1)

ASSEMBLY (all from the back)
1. Display (4), tab first into the lug pocket, PCB on the housing step.
2. Motor (6) into the ring on the lug skin; button (7) into its wall hole.
3. Midframe (2) with XIAO (5, USB-C into the wall hole), battery (8) and battery switch (10).
4. Back cover (3): 4 rim + 2 lug hooks snap into the housing grooves.
5. Screen protector (9) on the clean, dry front face.
Overall Ø42.2 x 65.8 x 13.74.
Section fill: grey = purchased parts.

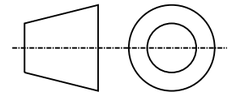


SECTION B-B
(4:1)

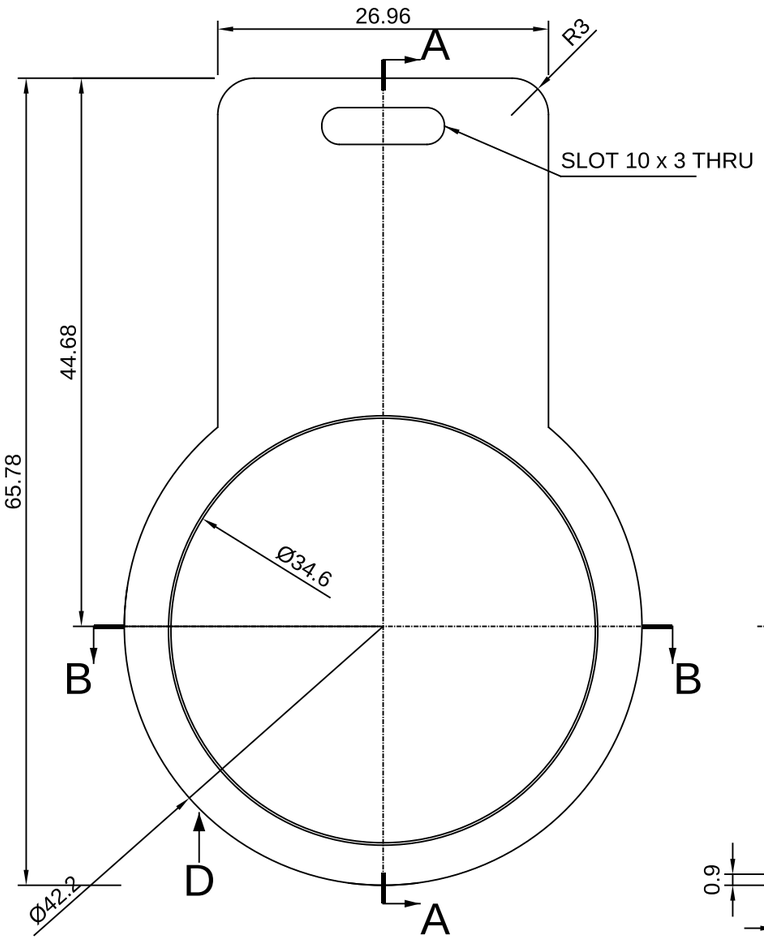
ITEM	PART	QTY	MATERIAL / NOTE
1	Front housing	1	PETG, drawing MC-M-002
2	Midframe	1	PETG, drawing MC-M-003
3	Back cover	1	PETG, drawing MC-M-004
4	GC9A01 1.28in round TFT	1	Teyleten Robot module, 7-pin header removed
5	XIAO nRF52840 Sense	1	Seeed Studio; header pins in the midframe rails
6	Vibration motor 10 x 3	1	Coin type, adhesive, in the lug skin ring
7	Tact switch 3 x 6	1	4.3 high; power + menu button
8	LiPo 402030	1	3.7 V ~200 mAh, 20 x 30 x 4, beside the XIAO
9	Round screen protector	1	Ø38-40, bonded on the front face
10	Slide switch EG1390	1	Battery switch, on the midframe (MC-M-003)
11	Male pin header 2 x 7, 2.54	1	On the XIAO, spacer off, 4.3 behind the PCB

PROJECT
MysteryChain case

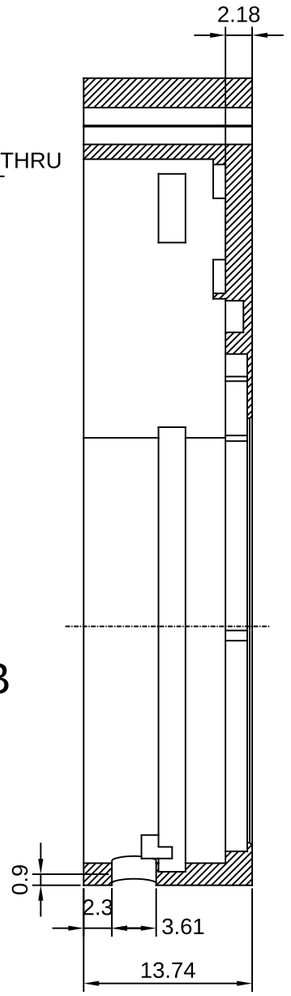
TITLE
ASSEMBLY

PROJECTION

FIRST ANGLE (ISO E)

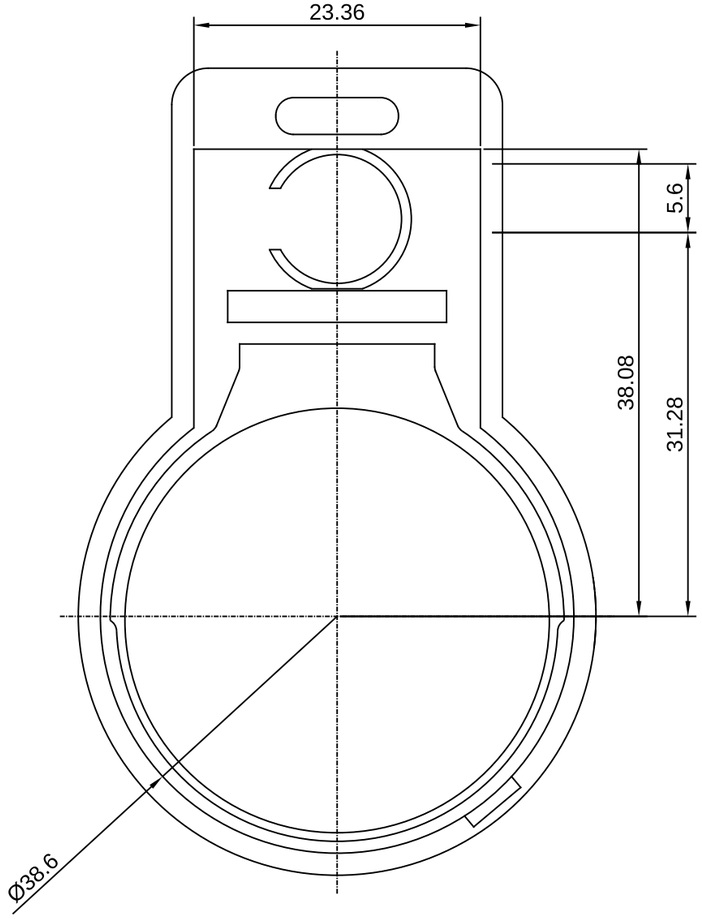
DRAWN	DATE	REVISION	DRAWING NO.
O. Eren	2026-10-24	v7.4	MC-M-001
MATERIAL	SCALE	UNITS / GEN. TOL.	SHEET
PETG	2:1	mm / ±0.2	1 / 4



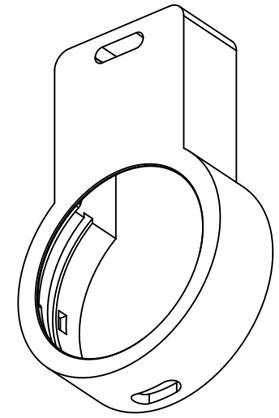
FRONT



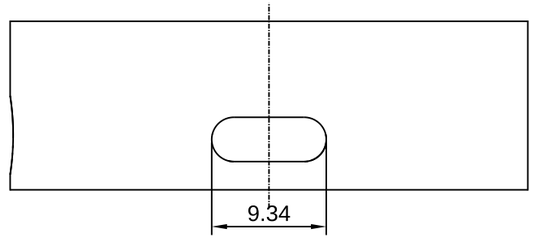
SECTION A-A



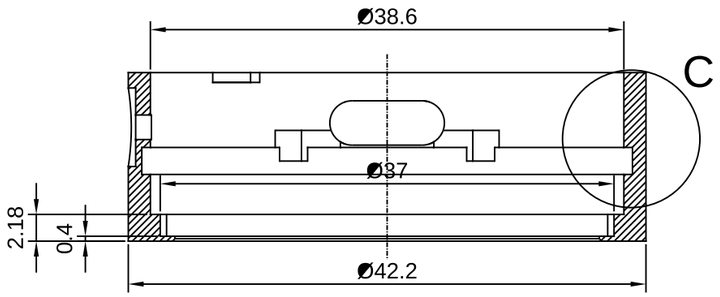
REAR



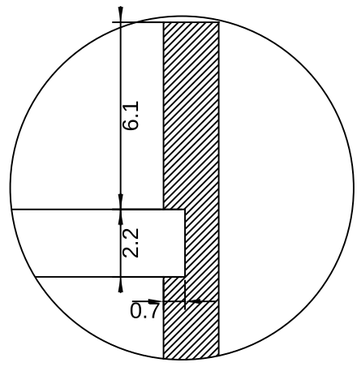
ISOMETRIC (1:1)



VIEW D



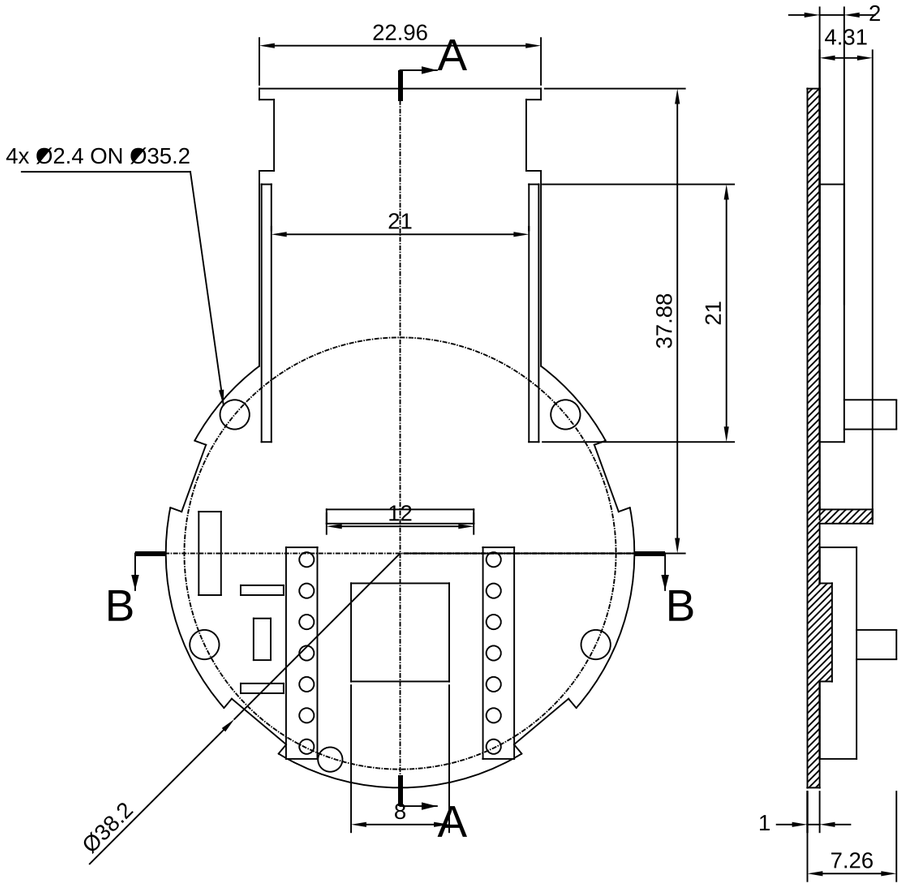
SECTION B-B



DETAIL C
(5:1)

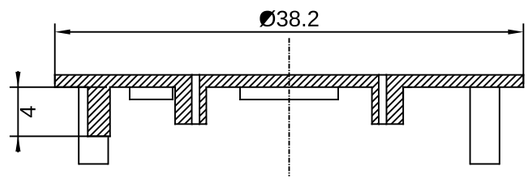
NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Window Ø34.6 (0.4 lip over the glass edge), centred 0.335 below the body axis (toward USB-C).
5. Hook groove R20.0 x z 6.1-8.3 (detail C); lug side grooves 0.7 deep, y 31.3-36.9. Pry notch 5 x 0.8 at 230 deg.
6. USB-C hole 9.34 x 3.61 slot (port shape, view D); the XIAO's USB end sits in a wall pocket (PCB outline + 0.25). Button hole 3.6 x 2.0, Ø6.4 dimple (-X).
7. LCD recess Ø36 (USB half), R18.5 on the tab half, both from the window centre; inner corners to the neck pocket R1.0.
8. LCD neck / FPC pocket: neck outline + 1.0 all round. Motor ring Ø10.5 x 1.0 on the lug skin.

PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: FRONT HOUSING			
DRAWN: O. Eren	DATE: 2026-10-24	REVISION: v7.4	DRAWING NO.: MC-M-002
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 2 / 4

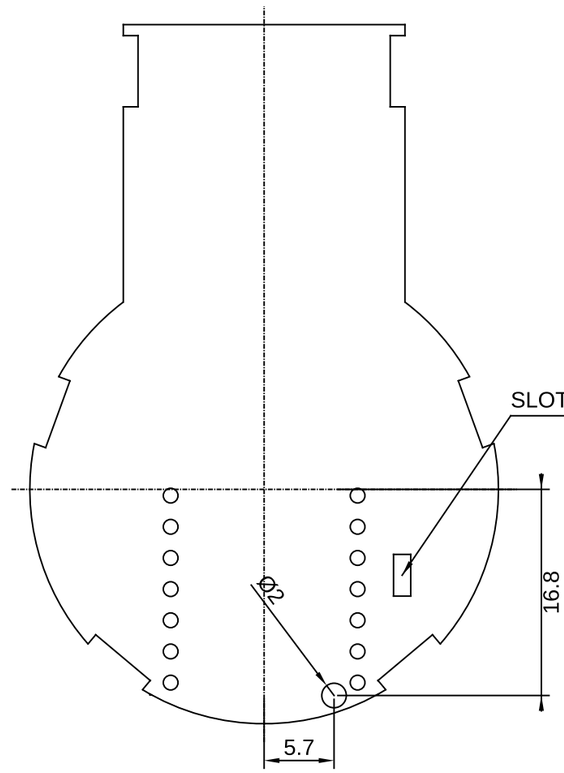


FRONT

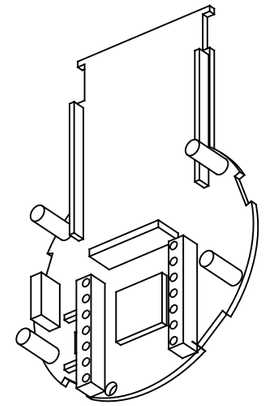
SECTION A-A



SECTION B-B



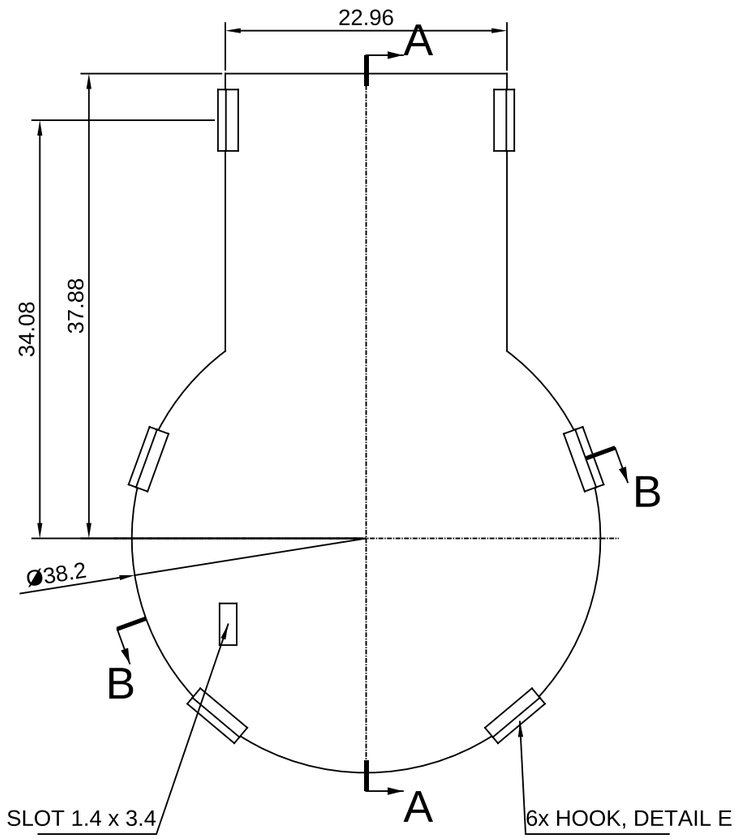
REAR



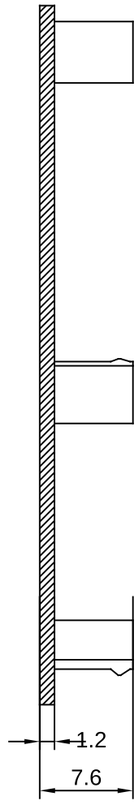
ISOMETRIC (1.2:1)

NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Display clamp legs 4x Ø2.4 at 40/140/205/335 deg, 1.5 short of the display PCB (tape).
5. The lug plate sets the orientation (no key tabs). Hook notches: 4 on the rim at 20/160/230/310 deg, 2 on the lug sides.
6. XIAO pin rails x = +-6.75..+-9.3, 2 x 7 holes Ø1.2 at x = +-7.62, pitch 2.54: the header pins press in and carry the USB-C force (tune PIN_HOLE_D). Tape pad 8 x 8 for 0.5 mm tape; stop rib 12 wide at the antenna end.
7. Battery side ribs 0.8 thick x 2 high. Battery switch EG1390 between two 0.8 end stops beside the XIAO, actuator through the 1.4 x 3.4 slot.
8. Reset hole Ø2 (pin, cover off). Print plate down, no supports.

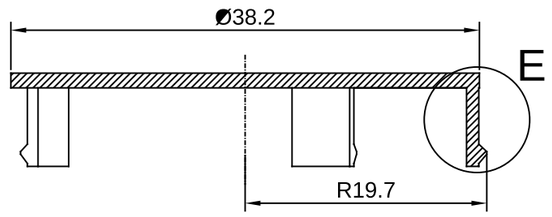
PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: MIDFRAME			
DRAWN: O. Eren	DATE: 2026-10-24	REVISION: v7.4	DRAWING NO.: MC-M-003
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 3 / 4



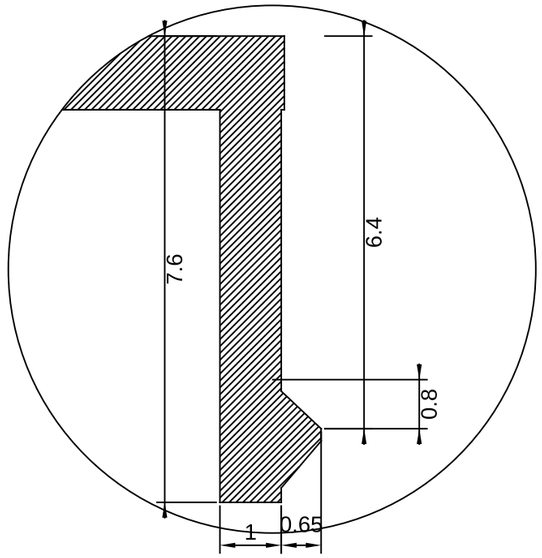
INSIDE (FROM FRONT)



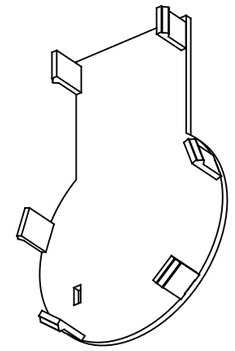
SECTION A-A



SECTION B-B (20 deg HOOK)



DETAIL E
(10:1)



ISOMETRIC (1:1)

NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Snap hooks 6x 5 wide: 4 on the rim at 20/160/230/310 deg, 2 on the lug edges at y = 34.1; profile per detail E.
5. Retention ramp 45 deg: the cover is removable (pry at the housing notch, 230 deg). The housing groove edge sits 0.08 into the ramp, so the hooks pull the cover in.
6. Battery switch slot 1.4 x 3.4 thru (tape over it).
7. Print plate down, hooks up, no supports.

PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: BACK COVER			
DRAWN: O. Eren	DATE: 2026-10-24	REVISION: v7.4	DRAWING NO.: MC-M-004
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 4 / 4